

DiagnosticIndex enrichment in BayPass outlier association regions

Formica hybrid zone, PC1/PC2 genotype–environment association

R 4.6.1 · data.table 1.18.2.1 · ggplot2 4.0.3

S1 Marker clusters and outlier region definition

Markers were grouped into linkage-disequilibrium (LD) clusters using a two-stage complexity-reduction procedure (LDscnR). Stage 1 applied complete-linkage clustering of pairwise LD (`ld_complexity_reduction()`). Stage 2 applied a distance-restricted, quality-gated average-linkage dynamic cut (`ld_prune_and_eMLG()`; `ld_w_threshold = 0.025`, `min_n_loci_flag = 5`, `min_n_loci_eMLG = 5`, `distance_threshold = 5×105 bp`). A cluster received a consensus expected multi-locus genotype (eMLG) if it contained at least 5 member markers (`has_eMLG = TRUE`). All analyses below are restricted to markers belonging to a `has_eMLG` cluster.

Per-marker association Bayes factors (BF(dB)) were obtained from BayPass for eight configurations, factorial in population set (Åland included, `with_aland`; Åland excluded, `aland_excluded`), principal component (PC1, PC2), and covariance-matrix treatment (no Ω correction, `no0mega`; Ω estimated from pruned markers, `with0mega`).

Within each configuration, an **outlier cluster** was defined as a `has_eMLG` cluster containing ≥ 10 individually significant member markers ($\text{BF(dB)} \geq 20$). This threshold was chosen to exclude clusters in which significance is driven by only a small minority of otherwise LD-linked members, which are more likely to reflect false positives (cf. companion LDscnR methods paper). Outlier-cluster membership was determined independently per configuration, yielding eight outlier sets. Markers in `has_eMLG` clusters not meeting this criterion, within the same configuration, form the **background** set for that configuration.

S2 DiagnosticIndex

DiagnosticIndex is DIEM's per-marker statistic of ancestry informativeness, assigned during genotype parsing. A marker was classified as ancestry-diagnostic if `DiagnosticIndex > -25`,

the threshold used elsewhere in this pipeline. Across all `n = 1,114,340` genotyped markers, 4.49% meet this criterion (genome-wide baseline).

S3 Statistical analysis

Cluster-size confound

The outlier definition (§S1) requires `n_loci ≥ 10` by construction, so outlier clusters are size-biased relative to the full background by design. Cluster size is independently associated with DiagnosticIndex genome-wide, prior to and independent of any BayPass result: a marker-level logistic regression of ancestry-diagnostic status on `log(n_loci)` alone gives OR = 1.135 per doubling of cluster size ($p < 10^{-300}$; Spearman $r = 0.23$ between cluster size and the fraction of diagnostic members). This is consistent with larger non-recombining regions being more likely to harbor multiple ancestry-informative loci, plausibly including Dobzhansky–Muller incompatibilities. Consequently, part of any naive outlier-versus-background enrichment could reflect cluster size rather than BayPass detecting genuine, size-independent signal.

Restricting the background to clusters of comparable minimum size (`n_loci ≥ 10`) does not resolve this: in a representative configuration, outlier clusters remained substantially larger than this size-floored background (median 147 vs. 15 loci; mean 743 vs. 28). A minimum-size floor alone was therefore judged inadequate, and cluster size was instead modeled continuously.

Enrichment tests

Two tests were computed per configuration, applied to markers in `has_eMLG` clusters only:

1. **Naive (unadjusted).** A 2×2 Fisher’s exact test on ancestry-diagnostic status (yes/no) by outlier status (outlier cluster / background), at marker resolution.
2. **Size-adjusted.** A marker-level logistic regression,
`DiagnosticIndex > -25 ~ outlier_status + log(n_loci)`, family binomial. The coefficient on `outlier_status` gives the odds ratio for ancestry-diagnostic enrichment attributable to outlier status after controlling for cluster size continuously. Confidence intervals are Wald-type (`confint.default`) rather than profile-likelihood, given sample sizes of order 10^5 – 10^6 markers per model; the two methods are expected to agree closely at this scale.

The size-adjusted test is treated as primary; the naive test is reported alongside it to show the magnitude of the size confound.

S4 Results

Table S1. DiagnosticIndex enrichment in outlier clusters, by configuration. pct_outlier/pct_background: percentage of markers with DiagnosticIndex > -25 in the outlier and background sets, respectively (genome-wide baseline odds ratio with 95% CI. Significance: *** p<0.001, ** p<0.01, * p<0.05, n.s. not significant.

Population set	PC	Ω	Clusters	Markers	% outlier	% bg.	Naive OR (95% CI)	Ad
Åland incl.	PC1	no Ω	3	2,161	21.9%	7.9%	3.27 (2.94–3.62) ***	2.03
Åland incl.	PC1	with Ω	4	2,284	20.8%	7.9%	3.04 (2.74–3.37) ***	1.93
Åland incl.	PC2	no Ω	13	9,664	11.0%	7.9%	1.44 (1.35–1.53) ***	0.75
Åland incl.	PC2	with Ω	20	15,787	27.5%	7.4%	4.75 (4.57–4.92) ***	2.79
Åland excl.	PC1	no Ω	5	1,469	31.9%	7.9%	5.44 (4.86–6.08) ***	3.76
Åland excl.	PC1	with Ω	6	1,540	30.4%	7.9%	5.08 (4.55–5.67) ***	3.56
Åland excl.	PC2	no Ω	3	593	53.6%	7.9%	13.43 (11.39–15.84) ***	10.31 (9.01–11.84) ***
Åland excl.	PC2	with Ω	4	3,776	8.4%	8.0%	1.06 (0.94–1.19) n.s.	0.54

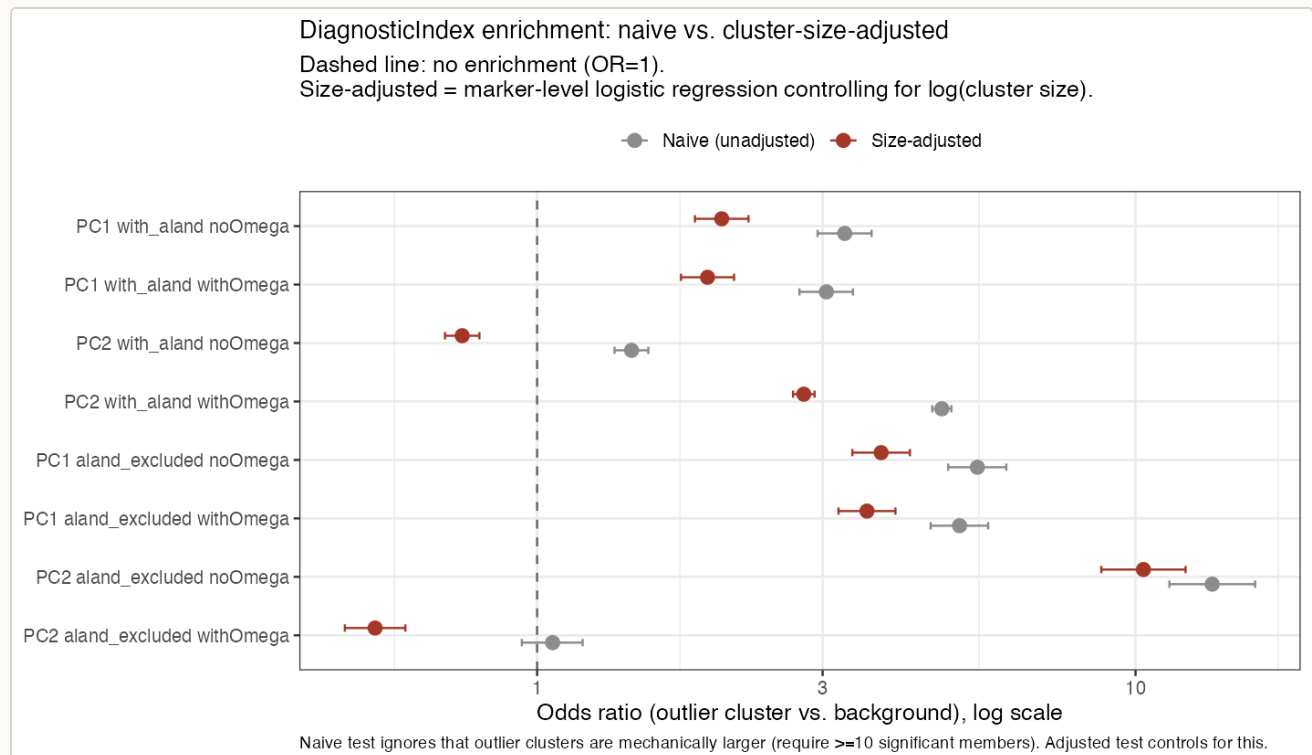


Figure S1. Naive (unadjusted) and size-adjusted odds ratios for DiagnosticIndex enrichment in outlier clusters, all eight configurations. Points: OR estimate; whiskers: 95% CI (naive: Fisher exact; adjusted: Wald, from the

outlier_status coefficient of the size-adjusted logistic model). Dashed line: OR = 1 (no enrichment).

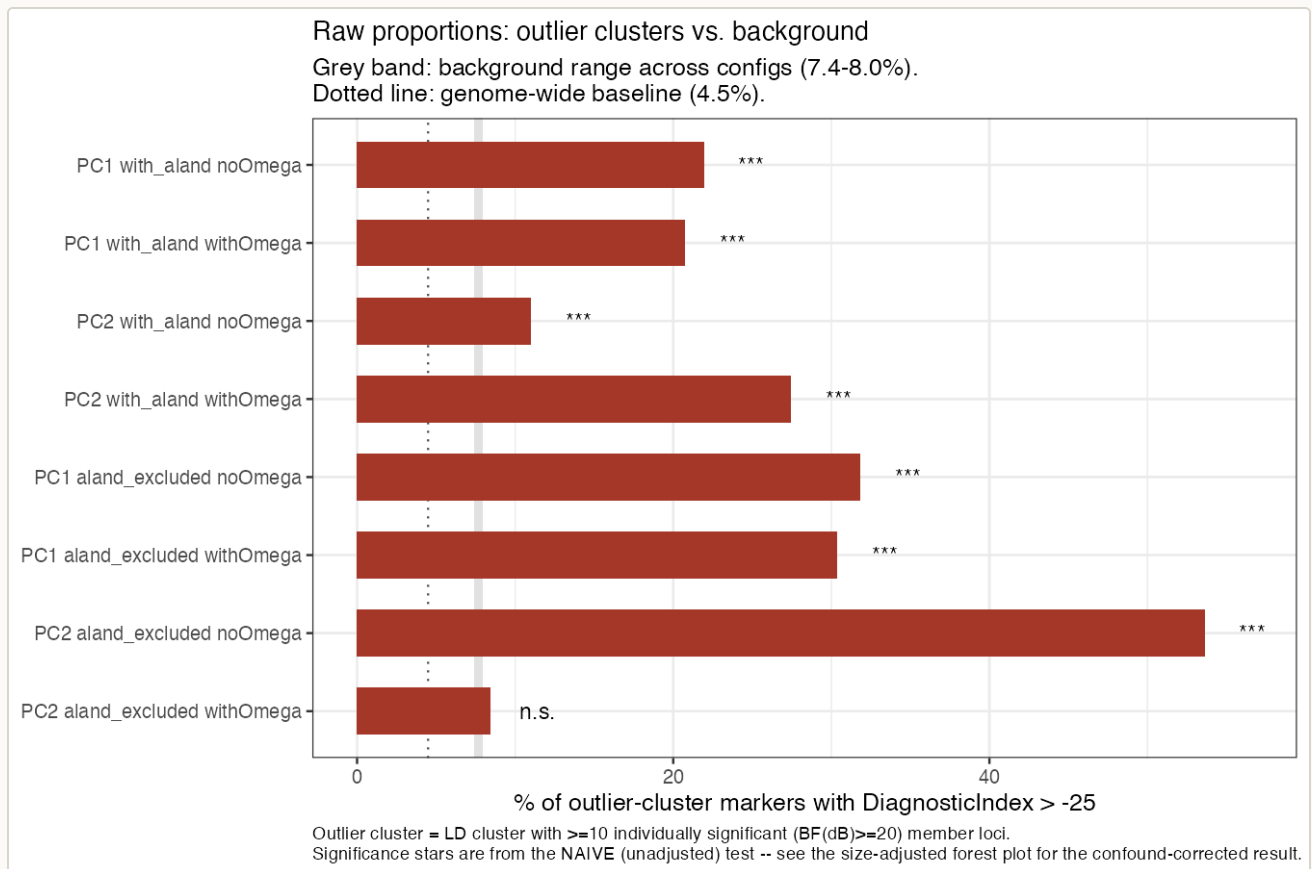


Figure S2. Percentage of outlier-cluster markers with DiagnosticIndex > -25, by configuration (unadjusted). Grey band: range of the background percentage across configurations (7.4–8.0%). Dotted line: genome-wide baseline (4.49%). Significance stars: naive Fisher test.

PC1 showed significant, size-adjusted enrichment in all four configurations in which it was tested (adjusted OR range 1.93–3.76, all $p < 10^{-35}$); adjustment attenuated but did not remove the effect. PC2 was significant under the naive test in all four configurations but did not retain a consistent direction after size adjustment: two configurations (with_aland/no0mega; aland_excluded/with0mega) reversed to significant depletion (adjusted OR < 1) after adjustment, while the remaining two (with_aland/with0mega; aland_excluded/no0mega) remained significantly enriched.

S5 Reproducibility

Analysis script: `R/diagnostic_index_enrichment.R`. Primary model, fit independently per configuration:

```
m <- glm(DiagnosticIndex > -25 ~ is_outlier + log(n_loci),
         data = marker_group, family = binomial)
```

```
ci <- confint.default(m, "is_outlier")
```

Input data: BayPass `summary_betai_reg.out` per configuration (joined to markers positionally via the `MRK` column, row-count- and order-checked against `map_hyb_005` before use); eMLG cluster assignments from `ld_prune_and_eMLG()` output (`eMLG_5loci_0025.rds`); DiagnosticIndex from `map_hyb_005` (DIEM parsing output).

Formica hybrid zone project · internal supplementary methods draft